

Yash Verma

Software Engineer · Java · Spring Boot · Distributed Systems · Microservices

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Summary

Software Engineer with 3+ years of experience building high-availability, low-latency Java backend systems and **Spring Boot** microservices at Bharti Airtel, serving platforms with **8.5M+** daily active users. Delivered measurable outcomes including **96%** reduction in validation turnaround, **60%** fewer deployment errors, and **99.97%** API success rates across distributed, fault-tolerant architectures. Seeking to bring production-grade distributed systems experience to large-scale engineering challenges at a product-first organization.

Skills

Languages: Java (8/11/17), SQL, JavaScript

Backend: Spring Boot, Spring MVC, Spring Security, Spring Cloud, Hibernate, JPA, Spring Batch, RESTful APIs, Microservices

Architecture: Distributed Systems, System Design, High Availability, Fault Tolerance, Scalability, Concurrency, Multithreading, Low-Latency Design, Event-Driven Architecture, Caching

Messaging & Streaming: Apache Kafka, RabbitMQ, Solace

Security: Spring Security, OAuth2, JWT, RBAC

Cloud & DevOps: AWS (EC2, S3, EKS, ECS, ECR, Lambda, CloudWatch, IAM), Docker, Kubernetes, Helm, Jenkins, CI/CD, Linux

Databases: Oracle, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, ChromaDB

Observability & Monitoring: Grafana, Kibana, Distributed Tracing

AI/ML: LLMs, RAG Pipelines, LangChain, LangGraph, Vector Databases

Testing: JUnit, Mockito, JMeter, VisualVM

Tools: Git, Maven, Agile/Scrum, Postman, Swagger, IntelliJ IDEA, GitHub Copilot

Core CS: Data Structures & Algorithms, Design Patterns, SDLC

Work Experience

Bharti Airtel

Software Engineer

Gurgaon, India

November 2022 – Present

ATOM (Order Management Platform)

Java, Spring Boot, Kafka, Redis

- **Owned end-to-end** lifecycle of high-availability backend microservices for broadband and postpaid order workflows — from design through deployment and **production incident response**
- Architected **Redis**-based distributed caching layer, reducing redundant API calls by **40%** and improving service latency under peak load; accelerated delivery with **GitHub Copilot**, cutting implementation time by **~30%**
- Designed and scaled **Spring Boot** microservices with **Kafka**-based asynchronous processing, improving **system throughput** and fault tolerance for high-volume order events
- Automated deployments and integrated testing into CI/CD pipelines via **Jenkins** and **Docker**, reducing deployment errors by **60%** and improving release stability
- Designed event-driven reconciliation workflows using **Drools Engine** and **Kafka**-based messaging, ensuring **100% Day-1** profile synchronization across distributed nodes

EPC Prepaid Revenue Assurance Automation Platform

Java, Spring Boot, React Native, REST APIs

- **Mentored** and guided **3 engineers** through system design discussions, code reviews, production debugging, and microservice development, improving engineering quality, **delivery velocity**, and adherence to backend best practices.
- **Architected** and led **end-to-end** delivery of a **full-stack automation platform** for Airtel's Prepaid Revenue Assurance team, eliminating a **multi-day** manual validation process across multiple prepaid billing systems
- Designed and developed a rule-based validation engine using **Strategy Design Pattern** and **Factory Design Pattern** in Java Spring Boot, enabling extensible validation workflows; leveraged **GitHub Copilot** to scaffold pattern classes and test cases, cutting boilerplate effort by **~25%**
- Designed a job scheduling and real-time status-tracking layer with automated alerting and discrepancy flagging, reducing validation turnaround from **5 days to under 30 minutes** and freeing analyst bandwidth by **80%+**
- Delivered a **React Native** frontend enabling Revenue Assurance analysts to trigger, monitor, and export structured mismatch reports from web and mobile, removing dependency on spreadsheet-based workflows

Device Nucleus (Provisioning & Diagnostics Platform)

Java, Spring Boot, REST APIs

- Designed and developed REST APIs and microservices for modem provisioning and diagnostics, integrated into the Airtel Thanks App serving **8M+** users

- Delivered APIs handling **3M+** requests/day with **99.97%** success rate using **auto-retry**, **failure recovery**, and observability via **Grafana** and **Kibana** dashboards
- Reduced customer service requests by **14%** through automated self-service troubleshooting flows

EPC Sync Validator (Backend Validation Platform)

Java, Spring Boot, REST APIs, Neo4j

- Designed and developed backend validation services for cross-system plan verification across **4 telecom billing systems**, executing **500+** automated checks with **98%** accuracy
- Architected **Neo4j** graph database integration to model and traverse complex billing relationships, enabling accurate dependency resolution across distributed billing nodes
- Delivered **96%** reduction in validation turnaround time (**5 days to 2 hours**), saving **160+** hours/month and freeing **3 FTE** for higher-value revenue assurance work

AI-Powered Router Diagnostics Platform

LangChain, Azure OpenAI (GPT-4o), RAG, Java, Spring Boot

- Built an **AI-powered router assistant** using **LangChain** agent orchestration with **Azure OpenAI GPT-4o**, enabling natural language interaction with **26+ router diagnostic tools** for **2,000+** field engineers
- Implemented **RAG (Retrieval-Augmented Generation)** pipeline with **text embeddings** and a **vector database** to ground LLM responses in router-specific knowledge, improving diagnostic accuracy and reducing hallucinations
- Designed **fault-tolerant** API orchestration with retry logic, achieving **92%** query resolution accuracy and reducing **Mean Time to Resolution (MTTR)** by **30%**
- Cut L1 support escalations by **25%** through automated AI-driven diagnostic workflows exposed via **Flask** REST API with a **React** chat frontend

Projects

CloudBox – Distributed Cloud File Storage System | *Java 21, Spring Boot 3, PostgreSQL, AWS S3, Redis, Kafka, Docker*

- Architected a **cloud-native** distributed file storage platform using **Java 21**, **Spring Boot**, **PostgreSQL**, and **AWS S3**, persisting binary objects in S3 while storing only file metadata in PostgreSQL through a clean **layered architecture**.
- Implemented secure **JWT**-based authentication and **role-based authorization** with **Spring Security**, exposing protected **REST APIs** for file upload, download, rename, delete, and metadata management documented with **OpenAPI/Swagger**.
- Improved metadata access performance by integrating **Redis caching** with targeted **cache invalidation** strategies for upload, rename, and delete operations, reducing repetitive database reads for frequently accessed file metadata.
- Designed an **event-driven** and **horizontally scalable** architecture using **Apache Kafka** for asynchronous file-upload workflows and **Nginx** load balancing for multiple stateless Spring Boot instances, improving extensibility and resilience through **decoupled background processing**.

Distributed Rate Limiter System (Token Bucket Based) | github.com/yashverma230600/distributed-rate-limiter

- Implemented Token Bucket algorithm in Java with **ReentrantLock**-based per-client locking – proven correct under **100** simultaneous threads with zero over-allowance
- Designed lock granularity at **per-client** level (not global) so independent clients never block each other under **concurrent load**
- Wrote **9 JUnit 5** tests, including a concurrency test firing 100 threads simultaneously via **CountDownLatch** – all passing
- Designed **Redis + Lua script** extension for distributed consistency across multiple instances – documented as next iteration

Education

GLA University

Bachelor of Technology – Computer Science and Engineering

Mathura, India

2019 – 2023

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Distributed Computing

Awards & Achievements

- **Ace Award & Rookie Award** – Recognized for delivery of innovative AI-integrated backend solutions at Bharti Airtel (2024)
- **LeetCode 200+ Problems Solved** – Strong problem-solving record across data structures, algorithms, and system design patterns
- **HackerRank 5-Star Problem Solver** – Ranked in top **10%** globally in Problem Solving certification